

Opinion Network Formation from Class Survey Responses

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GitHub Repository: [sarvesh2005takbhate/Titan_Mandal](https://github.com/sarvesh2005takbhate/Titan_Mandal)

Summary

We model a 60-statement Likert survey (Technology, Education, Ethics, Environment) as two networks: a **respondent network** (who thinks alike) and a **statement network** (which opinions travel together). Key findings:

- **A consensus class.** 79% of all answers are Agree/Strongly Agree. Environment and Ethics are near-unanimous; disagreement is concentrated in a handful of Education and Technology statements.
- **No distinct opinion camps.** Respondent communities ($Q = 0.299$) are only marginally stronger than in shuffled data (null $Q = 0.284 \pm 0.016$, $p = 0.22$) and change between runs ($ARI \approx 0.41$). Opinions vary along a continuum, not in blocs.
- **The clearest axis of disagreement:** support for regulation and traditional structure (T12, E02, E03) versus open, flexible learning (E04, E01, E09). All 4 negative statement correlations involve Education.
- **Themes cut across domains.** 57% of statement edges link different domains; the statement network forms 4 cross-domain clusters, including a *digital rights & AI regulation* cluster.

1. Dataset Documentation

Item	Value
Raw responses	96 respondents $\times$ 60 statements (15 per domain: T, E, S, V)
Encoding	Strongly Disagree = -2, Disagree = -1, Neutral = 0, Agree = +1, Strongly Agree = +2
Missing cells	542 of 5760 (9.4%): 505 blank, 37 "No Comments"
Excluded respondents	9 answered < 30 statements: 5 blank forms, 4 stopped after the Technology page
Analysed	87 respondents , 62 missing cells (1.2%)
Answer mix (analysed)	79% agree $\cdot$ 13% neutral $\cdot$ 7% disagree

- **Missingness is dropout, not item refusal.** All 505 blank cells come from respondents who abandoned the form (4 stopped exactly after the Technology block, consistent with a form paged by domain). Only the 37 "No Comments" are deliberate skips. Both are kept as NaN, never as Neutral; similarities use only statements both respondents answered. 2 partially complete respondents with ≥ 30 answers are retained.
- **Strong agreement bias.** Mean answers are high in every domain, so raw similarity would mainly measure *how much* someone agrees. This motivates centring (Section 2).
- **Response-style flags (retained, reported).** 7 respondents gave the same answer to $\geq 75\%$ of statements; #120 is a net dissenter (mean below zero); #119 answered mostly Neutral. These may be genuine views or low-effort answers.
- **Consistency check.** Near-duplicate statements in different domains correlate positively: T10-S05 $r = 0.46$; T02-E12 $r = 0.53$, suggesting answers are coherent.

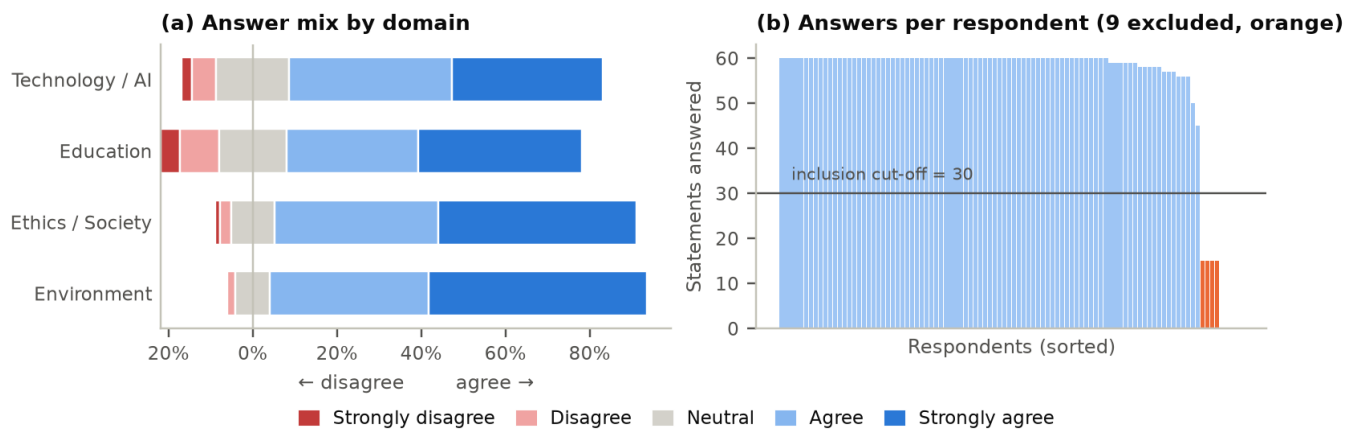


Figure 1: (a) Answer distribution by domain. (b) Statements answered per respondent; the 9 excluded respondents are orange.

2. Pipeline Followed

	Network 1: Respondents	Network 2: Statements
Nodes	87 respondents	60 statements
Similarity	Row-centred cosine over shared statements (≥ 5)	Pearson r across respondents (pairwise complete)
Edges	k-nearest neighbours, $k = 8$ (positive similarity only)	$\text{abs}(r) \geq 0.30$, signed (+ co-endorsement, – opposition)
Edge weight	similarity	$\text{abs}(r)$
Path distance	$1 / \text{similarity}$	$1 / r$, positive edges only

- Clean.** Encode Likert answers, keep missing as NaN, and drop the 9 respondents with < 30 answers (otherwise they appear as isolated nodes and fake communities).
- Respondent similarity.** Subtract each respondent's mean answer, then take cosine similarity over shared statements. Centring also weakens the link between a node's degree and its agreement level ($r = 0.66 \rightarrow 0.42$; Figure 2).
- Sparsify.** Link each respondent to its 8 most similar peers. Similarity stays the edge weight; shortest paths use distance = $1/\text{similarity}$, so strong ties are short.
- Statement network.** Keep $\text{abs}(r) \geq 0.30$ ($p \approx 0.005$ at $n = 87$). Negative edges are kept for interpretation but excluded from path metrics and community detection, since opposition is not proximity.
- Analyse.** Louvain communities tested against 50 shuffled datasets and for stability; centrality; η^2 (variance explained by community); a split index for polarization; PCA and Cronbach's α for domain structure; sensitivity to k and threshold.

Design choices and why

Each choice was compared with the obvious alternative on this dataset. The last column gives the evidence.

Decision	Chosen	Instead of	Why / benefit (evidence from this data)
Missing answers	NaN, pairwise-complete	Code as Neutral	A blank is not an opinion. Coding blanks as Neutral would raise the Neutral share from 13% to 21% and invent 505 answers.
Incomplete forms	Exclude < 30 answers	Impute	Imputing 45–60 of 60 answers fabricates a profile; kept as-is, the 5 blank forms become isolated nodes and fake communities.
Respondent similarity	Row-centred cosine	Raw cosine, Euclidean	Compares <i>which</i> statements a person favours, not how much they agree: mean similarity $0.68 \rightarrow 0.26$. Euclidean distance also grows with the number of shared answers.

Decision	Chosen	Instead of	Why / benefit (evidence from this data)
Sparsification	k-NN, $k = 8$	Global similarity threshold	A threshold with the same edge count leaves 15 respondents isolated (16 components); k-NN gives every respondent $\geq k$ neighbours and is connected from $k = 2$. $k \approx \sqrt{n}$, mid-range of the tested 5–12.
Path cost	1 / similarity	1 – similarity, hops	Keeps strong ties short while penalising weak ties; betweenness ranks barely change with 1 – similarity ($p = 0.93$), so results do not hinge on it.
Communities	Louvain	Greedy, label propagation, Girvan–Newman	Fast, weighted modularity optimisation. Greedy modularity reaches similar Q (0.31, 5 groups) but a different split (ARI 0.44); label propagation finds 1 community — consistent with weak structure. Girvan–Newman is $O(m^2n)$ with no natural stopping point.
Significance	Shuffled-answer null model	Raw Q, configuration model	k-NN graphs are modular by construction (null $Q = 0.28$). Shuffling keeps every statement's answer distribution and the full pipeline, so it tests exactly whether <i>combinations</i> of opinions cluster.
Partition agreement	ARI	NMI	Chance-corrected: unrelated partitions of the same sizes give $ARI \approx 0.00$ but NMI 0.06, which rewards chance overlap.
Statement association	Signed Pearson r	Spearman; $abs(r)$	Spearman agrees closely ($r = 0.93$ over all pairs). Keeping the sign separates opposition from co-endorsement; $abs(r)$ would have merged the 4 Education tensions into clusters.
Edge threshold	$abs(r) \geq 0.30$	0.20 or 0.40	~9 chance edges (3%) vs ~115 at 0.20; 0.40 isolates 12 statements.
Centrality	Betweenness (on distance)	Degree, eigenvector	In k-NN every degree $\geq k$ by construction; eigenvector centrality tracks agreement level more ($r = 0.52$) than betweenness does ($r = 0.34$). Betweenness captures bridging.
Polarization	Split index	Variance	Variance ranks E03 #2 most polarizing although 70% reject it; the split index is high only when both camps are large and reads directly in %.
Community profile	η^2 per statement	Domain averages	Size-weighted 0–1 effect size, comparable across statements; domain averages mix opposing statements (Education $\alpha = 0.58$).
Domain validity	Cronbach's α , PCA	Assume domains coherent	Tests whether a domain average is meaningful; PCA separates general agreement (PC1) from substantive disagreement (PC2).

Reproducibility. One command (`python notebooks/full_pipeline.py`) regenerates every number, table and figure from the raw CSV with fixed seeds. Unit tests check the similarity, η^2 , split-index and filtering maths, and all results are exported to `results.json`.

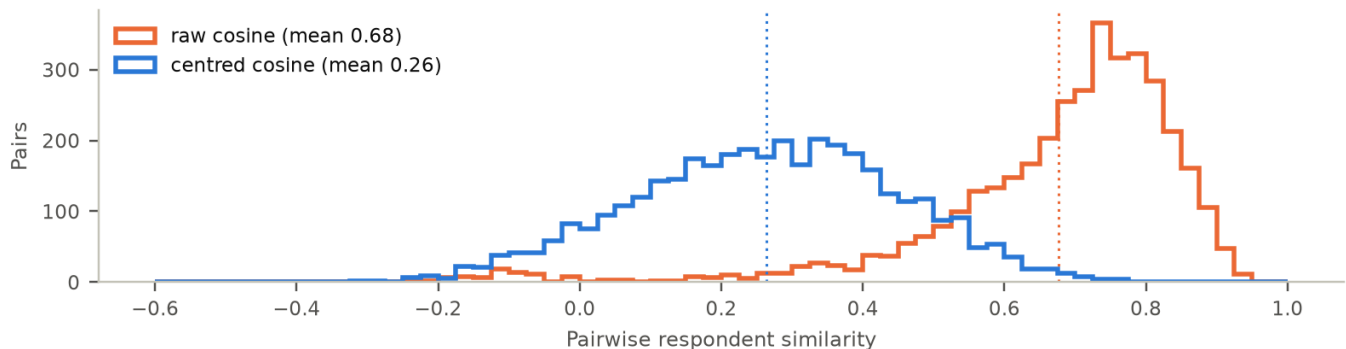


Figure 2: Pairwise respondent similarity before and after centring each respondent's answers.

3. Analysis and Visualizations

3.1 Network-level metrics

Metric	Respondent network	Statement network (positive edges)
Nodes	87	60
Edges	559	312
Density	0.149	0.176
Average degree	12.85	10.4
Weighted clustering	0.196	0.246
Clustering ÷ random-graph expectation	2.07	2.3
Connected components	1	7
Largest component	87	54
Avg. shortest path (hops)	2.06	2.1
Diameter (hops)	4	5
Avg. path length (distance units)	4.41	5.4

Paths are short in both networks (about 2 hops). Clustering is 2.1× (respondents) and 2.3× (statements) what a random graph of the same density would give, so similar opinions form local triangles rather than random links. After cleaning, the respondent network is a single component. In the statement network 6 statements have no positive link; see Section 3.3.

3.2 Respondent communities

- Louvain finds **5 communities** (sizes 21, 21, 17, 14, 14), $Q = 0.299$.
- **Null model:** networks from shuffled answers give $Q = 0.284 \pm 0.016$ ($z = 0.9$, $p = 0.22$). The observed structure is **not significantly stronger than chance**, because k-NN graphs are modular by construction.
- **Stability:** agreement with the main partition is $ARI = 0.47$ across Louvain seeds and 0.41 ± 0.11 when 10% of respondents are removed. Communities are soft groupings.
- **What differs between them:** community membership explains most variance for T07, E04, S08, T09 ($\eta^2 = 0.28\text{--}0.32$; mean η^2 by domain: T 0.16, E 0.11, S 0.12, V 0.11). No community disagrees with the class overall; they differ in emphasis.
- The groups line up with the opinion axis of Section 3.5: community C1 leans towards open learning (mean PC2 score -1.41) and C4 towards regulation and structure (+1.79); community explains $\eta^2 = 0.38$ of that axis.
- **Centrality:** the most central respondents by betweenness are #114, #49, #104. Their positions are structural (bridging similar-minded groups), not a measure of influence.

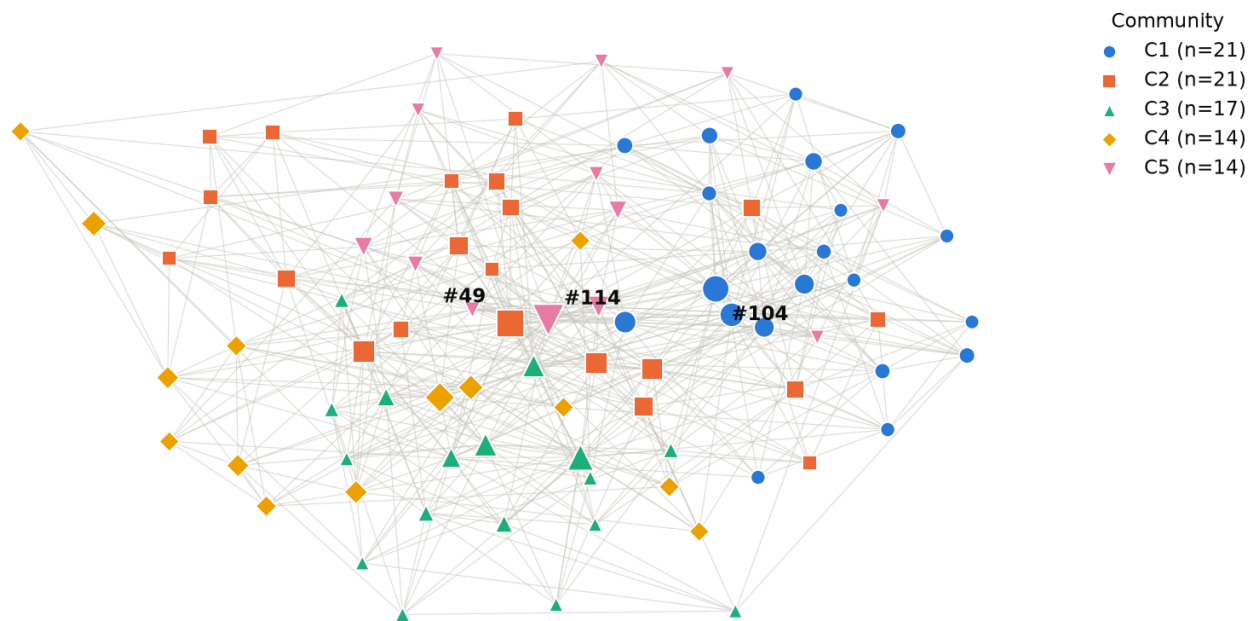


Figure 3: Respondent network ($k = 8$, centred cosine). Colour and shape show Louvain community; node size shows betweenness.

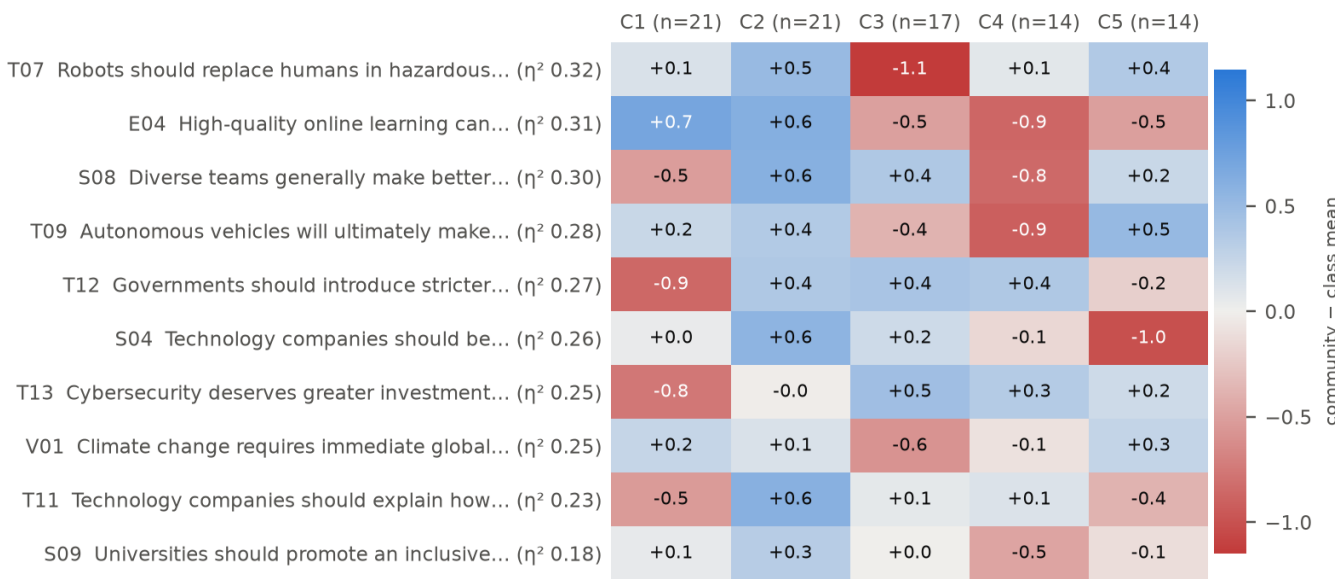


Figure 5: Community mean minus class mean on the ten statements that best separate communities (highest η^2). Blue = more agreement, red = less.

3.3 Statement network

- **Signs:** 312 positive and 4 negative edges. Every negative edge involves an Education statement:

Statement A	Statement B	r
E03 (Class attendance should be compulsory for all courses.)	E10 (Universities should prioritize innovation and problem-...)	-0.44
T03 (Students should disclose the use of AI in assignments...)	E04 (High-quality online learning can effectively...)	-0.32
E01 (Project-based learning develops deeper understanding...)	E02 (Traditional written examinations accurately measure a...)	-0.31
T06 (Most repetitive jobs will eventually be replaced by...)	E03 (Class attendance should be compulsory for all courses.)	-0.30

- **Cross-domain themes:** 57% of edges connect different domains. Louvain on positive edges gives 4 clusters ($Q = 0.21$), much more modular than the domain labels themselves ($Q = 0.13$; ARI with domains = 0.14):

Cluster	Theme	Statements
1	Environmental & social responsibility	S01, S03, S07, S08, S09, S13, S14, S15, T05, T07, V01, V02, V03, V04, V10, V11, V12, V13
2	AI-ready, industry-linked education	E07, E08, E10, E11, E12, E14, E15, S11, T01, T02, T03, T04, T06, V09, V14
3	Active learning & civic action	E01, E04, E05, E06, E09, S02, S04, S06, S10, S12, V05, V06, V07, V08, V15
4	Digital rights & AI regulation	S05, T10, T11, T12, T13, T14

- **Unconnected statements:** E13, T08, T09, T15 have no edge at all. E13 (49% Neutral), T08 (35% Neutral), T09 (38% Neutral) are among the five most-Neutral statements in the survey: uncertainty, not conviction, dominates them. E02, E03 connect only through negative edges.
- **Bridges:** highest betweenness: S09 (Universities should promote an inclusive...), E10 (Universities should prioritize innovation...), S07 (Equal opportunities should be prioritized...). Divisive statements sit at the periphery (split index vs weighted degree $r = -0.53$).

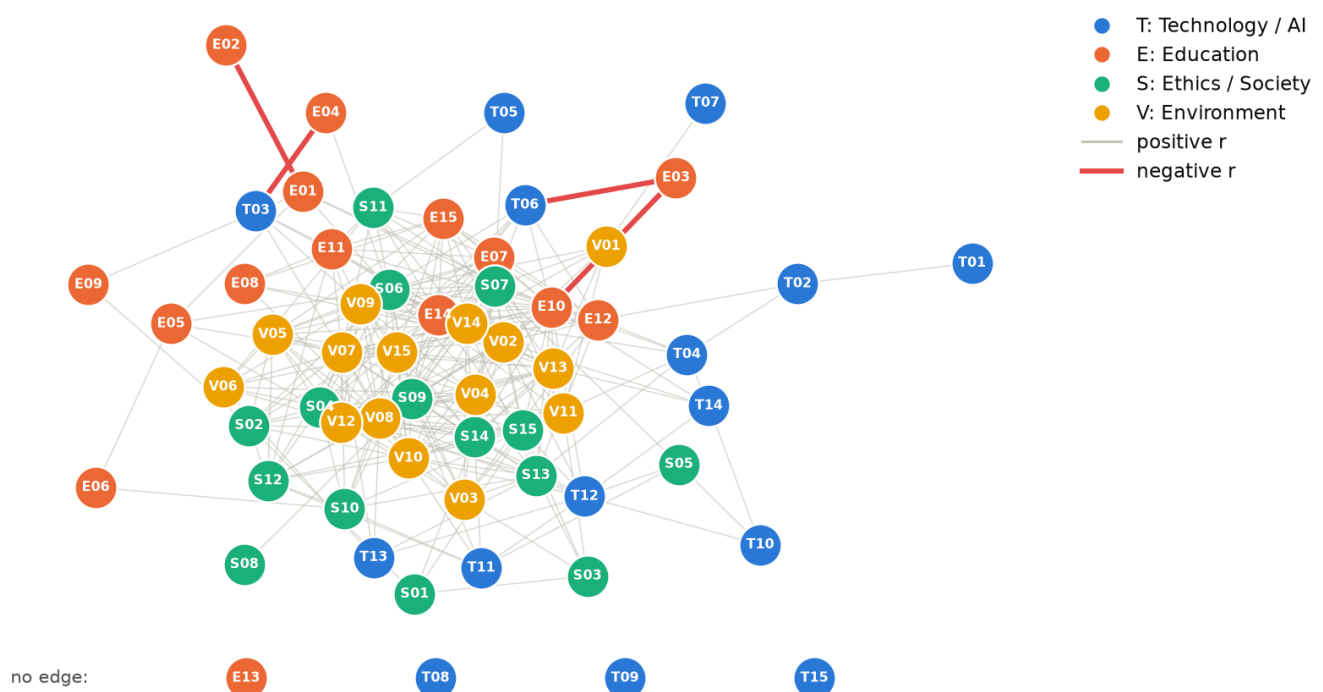


Figure 4: Statement network ($abs(r) \geq 0.30$). Colour shows domain (also the code letter); red edges are negative correlations; unconnected statements are shown along the bottom.

3.4 Polarization and consensus

Variance mixes a split class with a lopsided one, so we rank polarization by the **split index** = $\min(\% \text{ agree}, \% \text{ disagree})$, which is high only when both camps are large.

Group	Statement	Disagree	Neutral	Agree	Mean
Most split	E02 (Traditional written examinations accurately measure a...)	43%	25%	32%	-0.16
Most split	T08 (AI-assisted diagnosis should become routine in healthcare.)	28%	35%	37%	+0.07
Most split	E04 (High-quality online learning can effectively complement...)	28%	18%	54%	+0.44
Most split	T09 (Autonomous vehicles will ultimately make transportation...)	20%	38%	42%	+0.27
Most split	E13 (Universities should invest more resources in research than...)	13%	49%	38%	+0.34
Rejected	E03 (Class attendance should be compulsory for all courses.)	70%	17%	13%	-0.94
Consensus	E15 (Continuous learning and skill development are essential...)	0%	2%	98%	+1.71
Consensus	V13 (Companies should be held accountable for the environmental...)	1%	1%	98%	+1.67
Consensus	S09 (Universities should promote an inclusive environment where...)	0%	5%	95%	+1.47
Consensus	V06 (Water conservation should be a priority in households,...)	0%	5%	95%	+1.51
Consensus	S05 (Individuals should have greater control over how their...)	0%	6%	94%	+1.63

- Only 3 statements have $\geq 20\%$ on both sides; the most divisive is E02 (Traditional written examinations accurately measure a...) (32% agree vs 43% disagree).
- E03 is a consensus, not a split:** 70% reject compulsory attendance. High variance alone would have mislabelled it as polarizing.

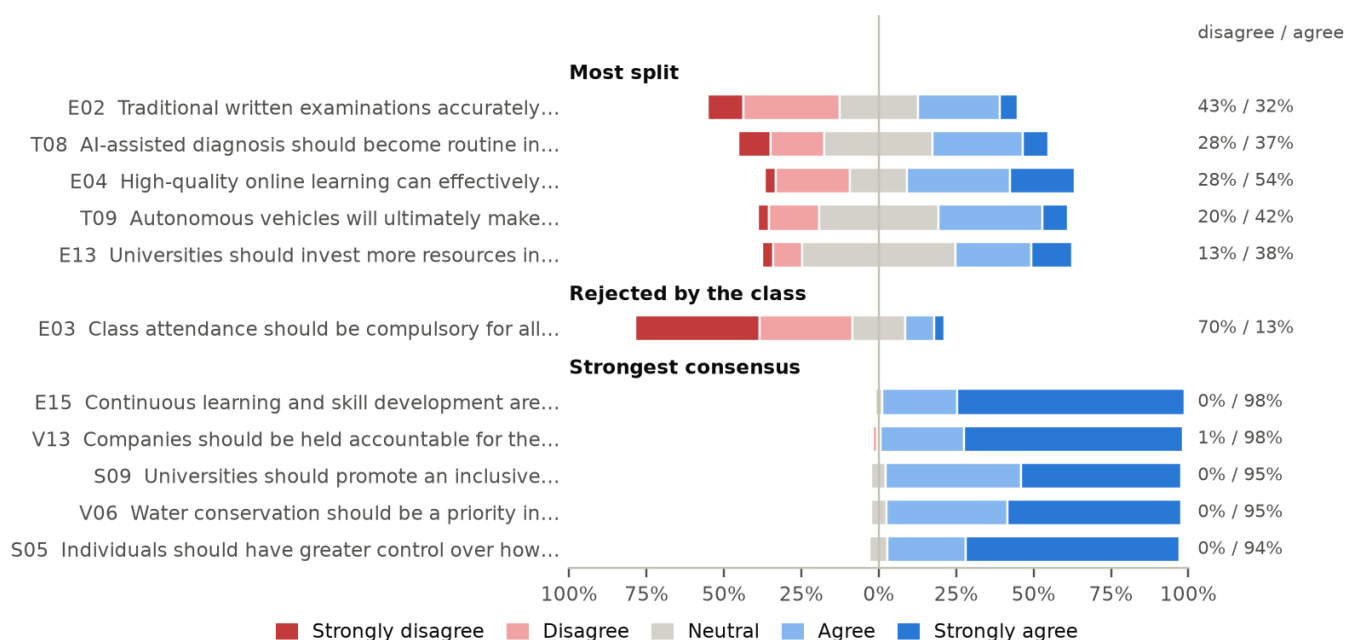


Figure 6: Answer distributions for the most split, rejected and consensus statements.

3.5 Domain structure and the clearest opinion axis

- Domain scores correlate positively.** Strongest S–V ($r = 0.70$), weakest T–E ($r = 0.29$).
- Reliability:** Cronbach's $\alpha = T$ 0.64, E 0.58, S 0.82, V 0.87. Technology / AI and Education fall below 0.70: they mix opposing positions (e.g. exams and attendance vs project-based and online learning), so a single domain average hides the real disagreement.
- PCA:** PC1 (17.2% of variance) is general agreement (95% of loadings share a sign; $r = 0.98$ with mean answer). **PC2 (6.2%) is the clearest substantive axis:** T12 (Governments should introduce stricter...), E02 (Traditional written examinations...), E03 (Class attendance should be compulsory...) versus E04 (High-quality online learning can...), E01 (Project-based learning develops deeper...), E09 (Collaborative learning is generally...). It is only slightly larger than PC3 (5.8%), so beyond general agreement opinions are weakly structured; PC2 stands out because it matches

the negative edges and the community differences.

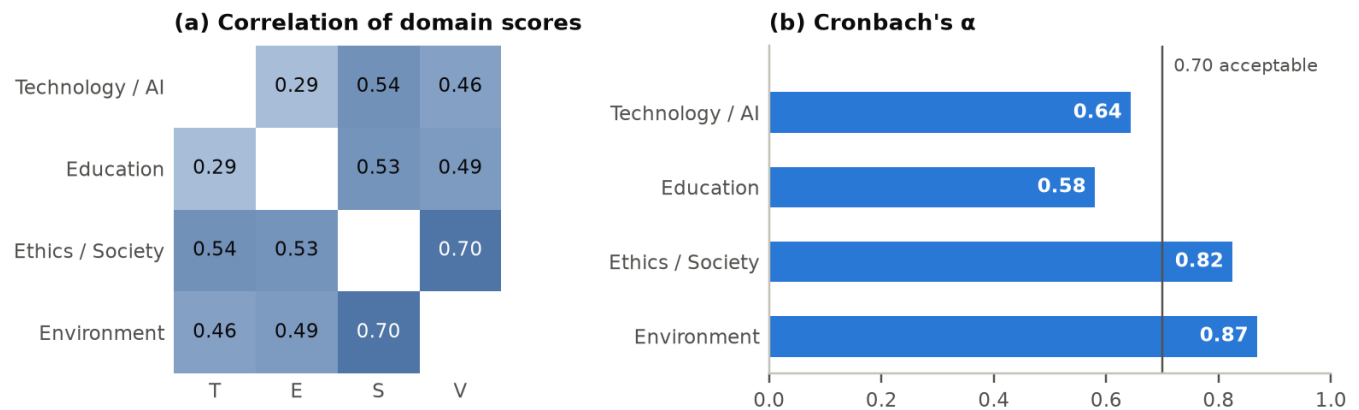


Figure 8: (a) Correlation between respondents' domain scores. (b) Internal consistency of each domain.

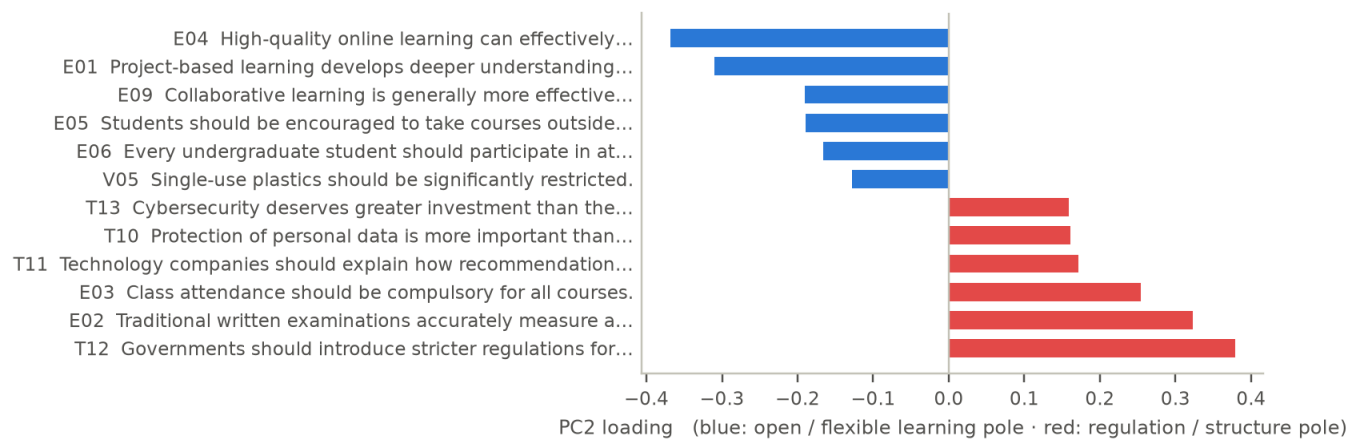


Figure 9: Statements with the largest loadings on the second principal component (6.2% of variance).

3.6 Robustness

- k:** modularity falls steadily from 0.36 ($k = 5$) to 0.23 ($k = 12$); partitions at other k agree only moderately with $k = 8$ (ARI 0.30–0.50), consistent with weak communities.
- Threshold:** edges drop from 709 at 0.20 (~115 expected chance edges) to 18 at 0.50. 0.30 keeps the network largely connected while limiting chance edges to ~3%.
- Similarity choice:** raw and centred cosine give different partitions (ARI = 0.34), another sign that respondent communities depend on modelling choices, whereas the statement-level findings are stable.

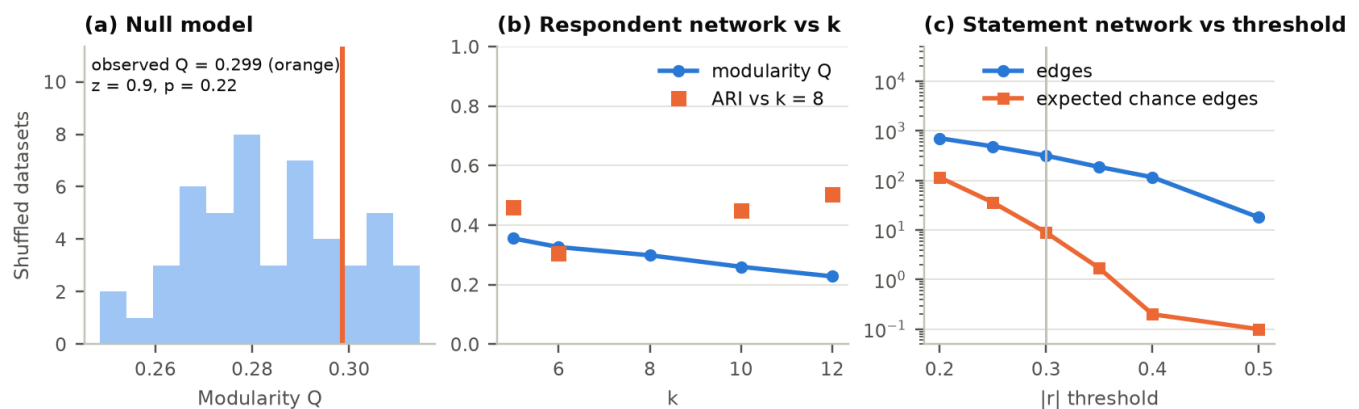


Figure 7: (a) Observed modularity vs shuffled-data null model. (b) Respondent network across k . (c) Statement network across thresholds (log scale).

4. Results and Discussion

1. **The class largely agrees.** Environmental and ethical statements (accountability, sustainability, inclusion) have the strongest consensus and the most internally consistent domains ($\alpha \geq 0.82$).
2. **Disagreement follows one recognisable axis.** Beneath the shared optimism, the main difference is attitude to control: regulating AI and protecting data, exams and compulsory attendance, versus online, project-based and collaborative learning. Education holds all 4 opposing edges and the most split statements.
3. **The network shows a continuum, not camps.** Communities are no stronger than in shuffled data and are unstable. Respondents differ by degree along the axis above, not by belonging to opposing groups. Averages alone could not show this; the null model makes it explicit.
4. **Opinions organise by theme, not survey section.** Most statement links cross domains, e.g. data privacy and AI regulation form one cluster (S05, T10, T11, T12, T13, T14). The four-domain survey layout does not match how respondents actually group issues.
5. **Uncertain topics stay disconnected.** Statements with many Neutral answers (AI diagnosis, autonomous vehicles, research vs infrastructure spending) do not correlate with anything, suggesting unformed rather than divided opinions.

Limitations

- Small, self-selected class sample ($n = 87$); strong agreement bias compresses the usable scale.
- Likert data are ordinal; Pearson r and cosine treat them as interval.
- k , threshold and similarity choices shape the respondent network; communities in particular are not robust.
- η^2 for communities is descriptive: communities are built from the same answers, so it is optimistic.
- Beyond general agreement the data are weakly structured (PC2 6.2% vs PC3 5.8% of variance).
- Edges are associations, not causal or social ties; centrality is not real-world influence.

Conclusion

The survey describes a broadly like-minded class whose meaningful disagreement is concentrated along one axis: structure and regulation versus openness and flexibility, expressed mostly through Education. Network analysis reveals this cross-domain structure. Significance and robustness checks show where the network is informative (statement associations) and where it is not (respondent communities).

5. Individual Contributions

- **Hrishikesh:** Data preparation, Likert encoding, missing-data handling, respondent similarity construction.
- **Sarvesh:** Network construction, Louvain community detection, centrality analysis, robustness experiments.
- **Shourya:** Visualization, statement-level analysis, report compilation and PDF generation.